INTRODUCTION TO MACHINE LEARNING

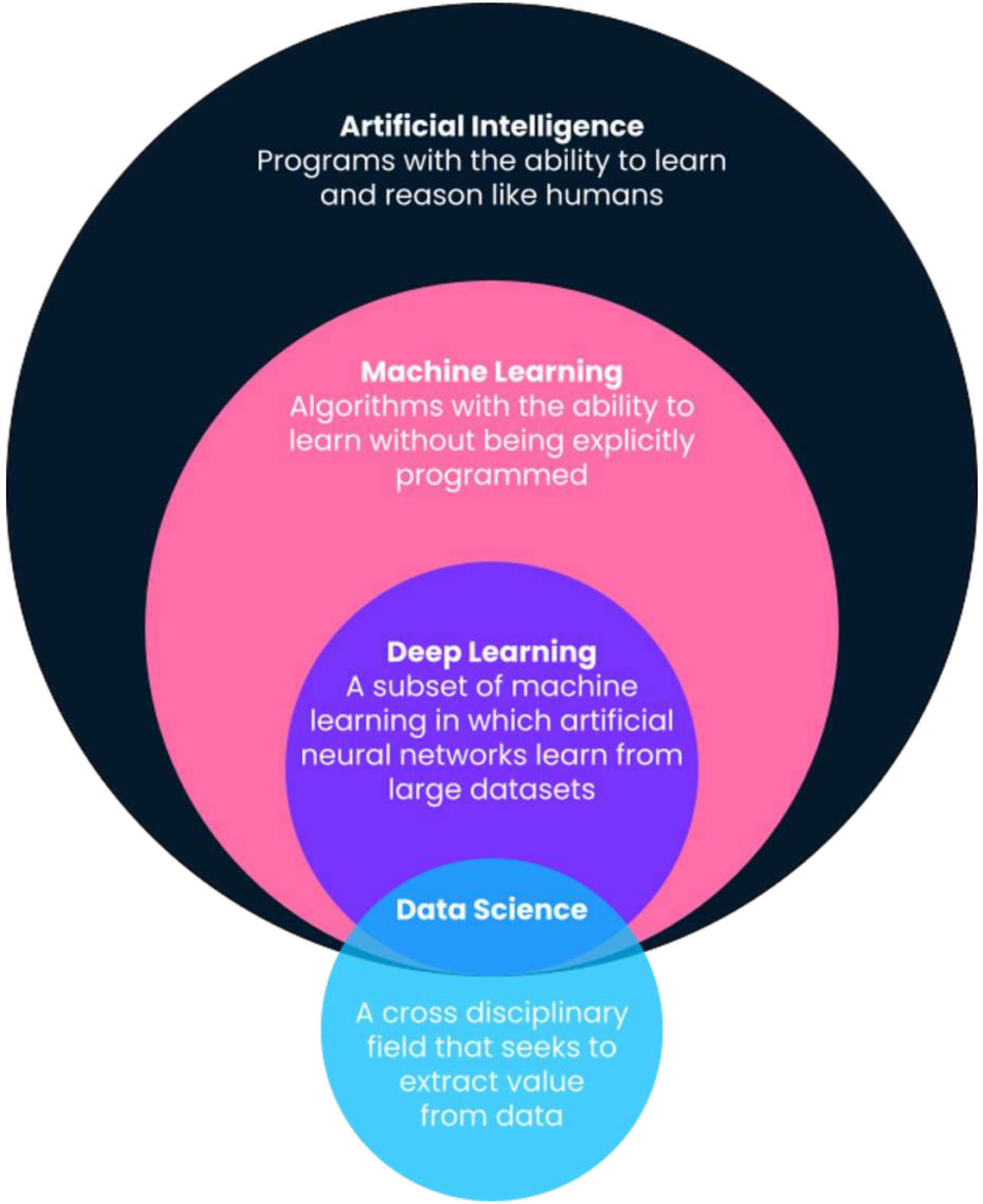
Eng. Maria Ayman

WHAT IS MACHINE LEARNING?

Machine learning is a subdomain of computer science that focuses on algorithm which help a computer learn from data without explicit programming (without telling it what to do!).

AI VS ML VS DS

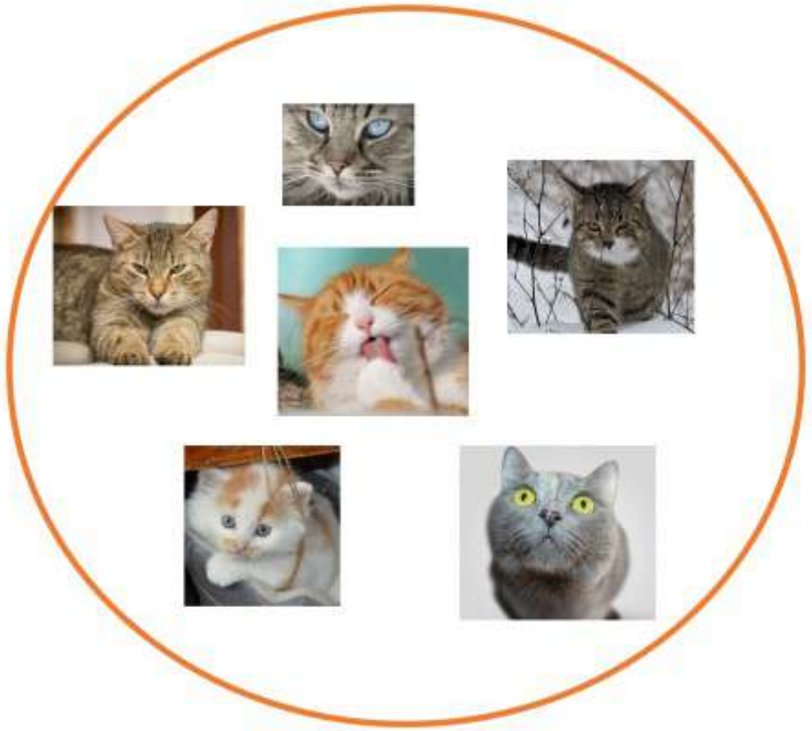
- **Artificial Intelligence:** is an area of computer science, where the goal is to enable computers and machines to perform human-like tasks and simulate human behavior.
- **Machine Learning:** is a subset of AI that tries to solve a specific problem and make predictions using data.
- **Data Science:** is a field that attempts to find patterns and draw insights from data (might use ML)


Artificial Intelligence
Programs with the ability to learn and reason like humans
Machine Learning
Algorithms with the ability to learn without being explicitly programmed
Deep Learning
A subset of machine learning in which artificial neural networks learn from large datasets
Data Science
A cross disciplinary field that seeks to extract value from data

TYPES OF MACHINE LEARNING

1. **Supervised learning:** uses labeled inputs (meaning the input has a corresponding output label) to train models and learn outputs.
2. **Unsupervised learning:** uses unlabeled data to learn about patterns in data.
3. **Reinforcement learning:** agent learning in interactive environment based on rewards and penalities.

SUPERVISED LEARNING

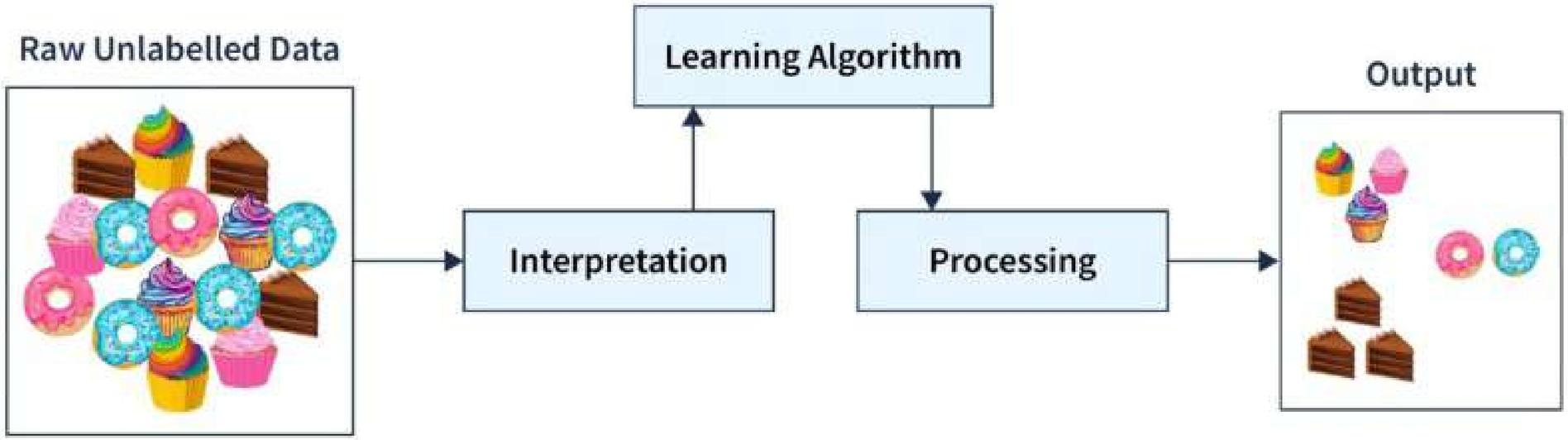


Cat



Dog

Unsupervised Machine Learning



REINFORCEMENT LEARNING



SUPERVISED LEARNING TASKS

1.Classification - predict discrete classes



Cat



Dog



Elephant

SUPERVISED LEARNING TASKS

1.Classification - predict discrete classes

Binary Classification

cat/ dog
positive/ negative
spam/ not spam

Multiclass Classification

cat/ dog/ elephant
orange/ apple/ banana
plant species

SUPERVISED LEARNING TASKS

2.Regression - predict continuous values







WHAT IS PYTHON?

1.Python is a high-level, interpreted programming language known for its simplicity and readability.

2.Python is an easy language to learn because of its simple syntax.

3.It is widely used in various domains such as web development, data science, artificial intelligence, machine learning, and more

WHY IS PYTHON?

1.Community and Documentation: Python has a large and active community of developers who contribute to its growth and development.

2.Extensive Libraries: Python has a vast ecosystem of libraries and frameworks for various purposes, allowing developers to leverage existing tools for their projects.

3.Versatility: Python can be used for a wide range of applications, from web development to scientific computing and machine learning.





Google colab

THANK YOU